EDS 223: Homework Assignment 3

Prioritizing Potential Aquaculture Along the West Coast

Ethan Mathews

2026-11-10

Load in Packages and Data

```
# Set up

# Load in libraries
library(tidyverse)
library(terra) # raster handling
library(sf) # vector handling
library(tmap)

# Load in sea surface data
sst_2008 <- rast("data/average_annual_sst_2008.tif")
sst_2009 <- rast("data/average_annual_sst_2009.tif")
sst_2010 <- rast("data/average_annual_sst_2010.tif")
sst_2011 <- rast("data/average_annual_sst_2011.tif")
sst_2012 <- rast("data/average_annual_sst_2012.tif")

# Load in depth data
depth <- rast("data/depth.tif")

# Load in EEZ data
eez <- st_read("data/wc_regions_clean.shp")
```

Stack and Process Data

```r
# Combine yearly raster layers into one raster
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)

# find mean SST from 2008-2012 and convert from K to C°
avg_sst_C <- mean(sst_stack, na.rm = TRUE) - 273.15

# Resample the depth data to match resolution of SST
depth_resample <- resample(depth, avg_sst_C, method = "bilinear")
```

Check layers have same resolution, extent, and CRS

```r
# Checking and converting depth to match SST resolution
if (all(res(avg_sst_C) == res(depth_resample))) {
  print("The resolutions match")
} else {
  warning("Update resolutions to match")
  res(depth_resample) <- res(avg_sst_C)
}
```

```
[1] "The resolutions match"
```

```r
# Checking and converting depth to match SST extent
if (ext(avg_sst_C) == ext(depth_resample)) {
  print("The extents match")
} else {
  warning("Update extents to match")
  ext(depth_resample) <- ext(avg_sst_C)
}
```

```
[1] "The extents match"
```

```r
# Checking and converting depth to match SST CRS
if (crs(depth_resample) == crs(avg_sst_C)) {
  print("The CRSs match")
} else {
  warning("Update CRSs to match")
  crs(depth_resample) <- crs(avg_sst_C)
}
```

```r
# Function to make maps of suitable aquaculture area

# Inputs:
#   min_sst, max_sst     - suitable sea surface temperature range (°C)
#   min_depth, max_depth - suitable depth range, as POSITIVE meters below
#   species_name         - species name for organization and the map title
#
# This function requires data prepared from potential_aquaculture.qmd:
#   avg_sst_C      - mean 2008-2012 SST in °C
#   depth_resample - depth cropped and resampled to match avg_sst_C
#   eez            - West Coast eez polygons, ensure the same CRS as rasters

suitable_zones <- function(
  min_sst,
  max_sst,
  min_depth,
  max_depth,
  species_name
) {
  # Reclassify SST by making matrix
  rcl_sst <- matrix(
    c(
      -Inf,
      min_sst,
      0,
      min_sst,
      max_sst,
      1,
      max_sst,
      Inf,
      0
    ),
    ncol = 3,
    byrow = TRUE
  )

  # Apply it to raster layer
  sst_zones <- classify(avg_sst_C, rcl = rcl_sst)

  # Reclassify depth by making matrix
  rcl_depth <- matrix(
    c(
```

```
    -Inf,
    -max_depth,
    0,
    -max_depth,
    -min_depth,
    1,
    -min_depth,
    Inf,
    0
  ),
  ncol = 3,
  byrow = TRUE
)

# Apply it to raster layer
depth_zones <- classify(depth_resample, rcl = rcl_depth)

# Find areas that satisfy both SST and depth
species_stack <- c(sst_zones, depth_zones)

species_suitable <- lapp(species_stack, fun = function(sst, depth) {
  sst * depth
})

# Rastertize EEZ
eez_raster <- rasterize(vect(eez), species_suitable, field = "rgn_id")

# Calculating area of each cell
cell_area <- cellSize(species_suitable, unit = "km", mask = TRUE)

# Keep area of suitable cells only; unsuitable (0) become NA
suitable_area <- mask(cell_area, species_suitable, maskvalues = 0)

# Select raster cells that fall in EEZ regions
suitable_area_eez <- zonal(
  suitable_area,
  eez_raster,
  fun = "sum",
  na.rm = TRUE
)

# Convert into DataFrame
```

```r
suitability_df <- as_tibble(suitable_area_eez) |>
  rename(suitable_area_km2 = area) # rename column name

# Join suitable areas to EEZ polygons
eez_map <- left_join(eez, suitability_df, by = "rgn_id")

# Make a map of West Coast EEZ's
tm_shape(eez_map) +

  # Add a basemap
  tm_basemap("OpenStreetMap") +

  # Customize legend
  tm_fill(
    fill_alpha = 0.75,
    fill = "suitable_area_km2",
    fill.legend = tm_legend(
      "Suitable Area (km²)",
      position = tm_pos_out("right", "center")
    )
  ) +

  # Customize border of polygons
  tm_borders(
    lwd = 0.5
  ) +

  # Add a scalebar
  tm_scalebar(
    position = c("left", "bottom"),
    text.size = 0.25
  ) +

  # Add a compass
  tm_compass(
    position = c("right", "top"),
    size = 1.5
  ) +

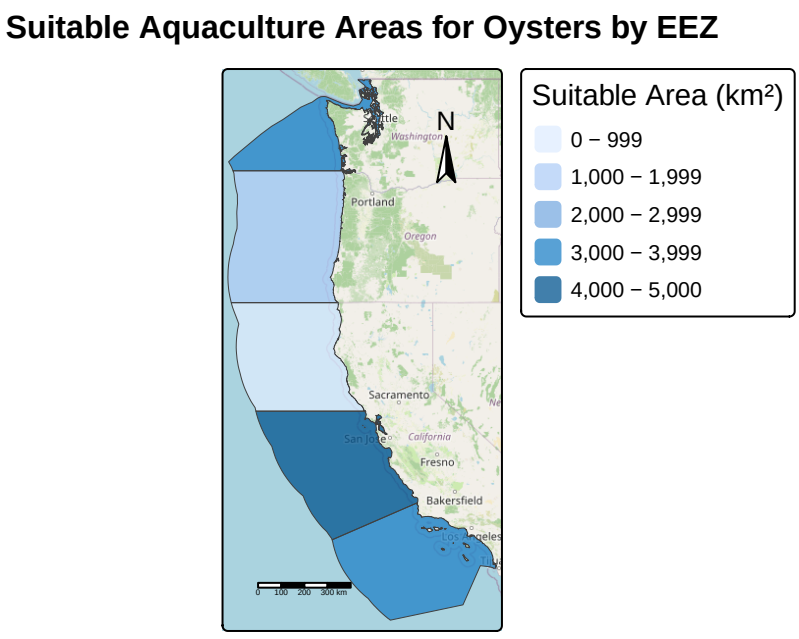
  # Add a title
  tm_title(
    paste("Suitable Aquaculture Areas for", species_name, "by EEZ"),
```

```
      fontface = "bold",
      size = 1
    ) +

    # Remove warning
    tm_options(component.autoscale = FALSE)
}
```

Maps

```
# Suitable zone map for oysters
suitable_zones(
  min_sst = 11,
  max_sst = 30,
  min_depth = 0,
  max_depth = 70,
  species_name = "Oysters"
)
```



```r
# Suitable zone map for red abalone
suitable_zones(
  min_sst = 8,
  max_sst = 18,
  min_depth = 0,
  max_depth = 24,
  species_name = "Abalone"
)
```

Suitable Aquaculture Areas for Abalone by EEZ

